



Effects of rearing environment and population origin on responses to repeated behavioural trials in cane toads (*Rhinella marina*)

Jodie Gruber^{a,*}, Martin J. Whiting^b, Gregory Brown^a, Richard Shine^a

^a School of Life and Environmental Sciences, The University of Sydney, Australia

^b Department of Biological Sciences, Macquarie University, New South Wales, Australia

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ABSTRACT

Behavioural responses to repeated trials in captivity can be driven by many factors including rearing environment, population of origin, habituation to captivity/trial conditions and an individual's behavioural type (e.g., bold versus shy). We tested the effect of rearing environment (captive raised common-garden versus wild-caught) and population origin (range-edge versus range-front) on the responses of invasive cane toads (*Rhinella marina*) to repeated exploration and risk-taking assays in captivity. We found that behavioural responses to identical assays performed on two occasions were complex and showed few consistent patterns based on rearing environment or population of origin. However, behavioural traits were repeatable across Trial Blocks when all sample populations were grouped together, indicating general consistency in individual toad behaviour across repeated behavioural assays. Our findings exemplify the complexity and unpredictability of behavioural responses and their effects on the repeatability and interpretation of behavioural traits across repeated behavioural assays in captivity. To meaningfully interpret the results from repeated behavioural assays, we need to consider how multiple factors may affect behavioural responses to these tests and importantly, how these responses may affect the repeatability of behavioural traits across time.

1. Introduction

Individuals often consistently differ in their responses to repeated exposure to behavioural trials and it is likely that multiple internal and external factors drive this variation (Bell and Peeke, 2012). Previous experience (Mason et al., 2013), coping style (whether individuals are proactive versus reactive; Wolf et al., 2008), sex (Ellenberg et al., 2009; Rodriguez-Prieto et al., 2011) and experiences during ontogeny (Zimmermann et al., 2001) all can affect behavioural responses. Habituation to captivity or trial conditions may also affect the behavioural responses of an individual across repeated trials. Habituation is a form of behavioural plasticity whereby an individual reduces its response to stimuli with repeated exposure over time (Rankin et al., 2009). Habituation may occur in response to factors such as the presence of predators, food availability, novel environments and human disturbance (e.g., anthropogenic noise and human-altered habitats) (Blumstein, 2016). Furthermore, an individual's behavioural type or 'personality' may affect its behavioural response across repeated trials, leading to variation over time. For example, Iberian wall lizards (*Podarcis hispanica*) that are more exploratory habituate faster (Rodriguez-Prieto et al., 2011), human extroverts have a faster habituation rate than introverts

(LaRowe et al., 2006), and 'calm' penguins recover from human disturbance more quickly than do aggressive conspecifics (Ellenberg et al., 2009).

Among-population divergence in behavioural traits such as exploration, risk-taking, sociability and neophilia (an attraction to novelty) occurs across range expansion gradients in several taxa including fishes (Chapman et al., 2011; Cote et al., 2011), birds (Atwell et al., 2012; Dingemans et al., 2003; Liebl and Martin, 2012, 2014), reptiles (Aragon et al., 2006; Chapple et al., 2012) and amphibians (Gruber et al., 2017a,c). Individuals with high levels of exploration, risk-taking and neophilia tend to be more common in range-edge populations as they are more likely to disperse from the range-core and exploit novel environments and resources than are shyer conspecifics (Cote et al., 2010; Rehage and Sih, 2004). Behavioural responses to repeated behavioural trials may also be expected to diverge among range-edge and range-core populations either due to links between behavioural types and habituation rates (e.g., bolder individuals habituate faster), or because of variation in other factors such as life-history or population demography across the invasion range.

Previous research has documented geographic divergence in exploration and risk-taking behavioural traits in cane toads (*Rhinella*

* Corresponding author at: The University of Sydney, A08, Heydon-Laurence Building, New South Wales 2006, Australia.
E-mail address: jodie.gruber@sydney.edu.au (J. Gruber).

marina) from populations across their Australian invasion range (Gruber et al., 2017a). By conducting single laboratory-based behavioural assays for each trait, Gruber et al. (2017a) revealed that invasion-front toads are more exploratory and have a higher propensity to take risks than do conspecifics from long-colonised populations. Behavioural assays testing the same traits in captive raised common-garden toads revealed the same pattern of behavioural divergence, suggesting a genetic component to these traits (Gruber et al., 2017c).

In this study, we extended the work of Gruber et al. (2017a,c) by testing how exploration and risk-taking behaviour changes with repeated exposure to identical behavioural trials in captivity as a function of (1) Population Origin; tested by comparing behavioural responses across repeated trials in wild-caught toads from long-colonised and invasion-front populations from Australia and long-colonised populations from two islands of Hawai'i (adding to research on Hawai'i toad behaviour; Gruber et al., 2017b), and (2) Rearing Environment (captive-raised versus wild-caught toads); tested by comparing the behavioural responses of Australian wild-caught toads with captive, common-garden raised toads of long-colonised and invasion-front ancestry.

We predicted that (1) wild-caught toads would be more exploratory and take more risks in Trial Block Two than One (i.e., toads would become bolder hence more active over time) as they adjusted to behavioural trials and captivity, (2) invasion-front toads would show less change in behaviour over time than would toads from long-colonised populations (invasion-front toads already have high levels of exploration and risk-taking in Trial Block One: Gruber et al. 2017a, c); and (3) common-garden raised toads would exhibit consistently high levels of exploration and risk-taking over time as they were raised in a captivity.

2. Materials and methods

We collected 68 adult cane toads comprising 34 individuals (17 male, 17 female) from each of two locations across their Australian invasion range: Cairns, Queensland (> 80 years since colonisation) and Oombulgurri, Western Australia (< 3 years since colonisation) at time of study. We collected 120 adult toads from two Hawai'i islands (both > 80 years since colonisation) comprising 60 (30 male, 30 female) from the island of Hawai'i (Big Island hereafter, to avoid confusion) and 60 from O'ahu (30 male, 30 female). We also used 34 (17 male, 17 female) common-garden raised toads from each of the two Australian populations of origin: long-colonised population (Queensland) and invasion-front population (Western Australia), which at the time of the study were approximately 18 months of age, and had been raised for their entire lives at the same site (The University of Sydney's Tropical Ecology Research Facility in the Northern Territory [12° 37' S, 131° 18' E]). Common-garden raised toads were selected from 13 clutches produced by parents originating from Western Australia and 16 clutches produced by parents originating from Queensland. Collection details, housing and breeding protocols for the common-garden raised toads are described in (Hudson et al., 2016). Specific details of collection procedures, housing, and trial methodologies are described in Gruber et al. (2017a,b,c) but brief descriptions appear below.

2.1. Housing

Toads were housed together in groups of two to three in 100 L plastic containers with mesh lids. Each tub had a wood-shaving substrate and contained two shelters (each shelter was large enough to hold three toads) and a large shallow water dish. Toads were fed live crickets or mealworms four times per week and were provided with fresh water *ad libitum*. Wild-caught toads and common-garden raised toads were housed in indoor captive conditions for one week before trials began. Housing and husbandry conditions were equally novel to both wild-caught and common-garden raised toads. Prior to the commencement

of this study, common-garden raised toads resided in large, open, outdoor enclosures where human interaction was rare and the toads foraged naturally.

2.2. General trial methods

All toads were exposed to two types of behavioural assay in the same sequence; an exploration trial followed by a risk-taking (emergence from a shelter into the test arena) trial. Each trial ran for 30 minutes. Four toads were tested concurrently, in separate identical arenas (hexagonal pens, 120 × 120 × 83 cm) with an open top to allow filming from above (see Gruber et al., 2017c for further details). After a 12-day period, each toad was subjected to a second replicate exposure to both behavioural tests. All trials were filmed using CCTV cameras under low-level red-lighting. We scored videos using Ethovision XT10 behavioural analysis software (ensuring that blind methods were used to score the videos).

2.3. Behavioural trials

2.3.1. Exploration trial

To test exploration, we measured time spent moving and rate of movement (quantified by the residual scores from a simple linear regression of total distance moved against total time spent moving). Toads were given five minutes rest under a shelter within the arena before the shelter was removed and trials began. An enclosed plastic take-away container (added for visual interest) and a shelter were also placed in the arena (at the opposite end of the arena to the starting point), giving toads the option to seek refuge during trials. This allowed us to distinguish bold exploratory behaviour from fear-driven escape behaviour (Réale et al., 2007).

2.3.2. Risk-taking (emergence) trial

To measure propensity to take risks, we recorded whether or not a toad emerged from its shelter, and latency to emerge (s). Emergence from the shelter and a shorter latency to emerge generally indicates higher risk-taking propensity in animals (Brodin et al., 2013; Cote et al., 2010). Toads began each trial with five minutes rest under a shelter with a covered entrance before the cover was removed and trials began. Two empty containers were placed at the opposite end of the arena to the shelter to provide visual interest in the otherwise white arena.

2.4. Statistical analysis

We used generalised linear mixed models in JMP 11.0 (SAS Institute, Cary, NC) to test whether changes in exploration and risk-taking behaviour between Trial Block One and Trial Block Two differed among toads from (1) different Rearing Environments (captive-born, common-garden raised toads with Queensland and Western Australian ancestry and wild-caught toads from Queensland and Western Australia) and (2) different Population Origins (wild-caught toads from long-colonised sites [Hawai'i and Queensland] versus wild-caught invasion-front toads from Western Australia). The effects included in the Rearing Environment model were Population Source (four-level categorical factor including common-garden Queensland, common-garden Western Australia, wild-caught Queensland and wild-caught Western Australia), Trial Block (two-level categorical factor) and their interaction. For the Population Origin model, we included population type (four-level categorical factor containing Queensland, Western Australia, Big Island and Oahu), Trial Block (two-level categorical factor) and their interaction. We ran separate models for the four behavioural traits. For each model, we used a single factor with four levels, allowing us to directly compare our four groups of interest (Queensland, Western Australia, Big Island and Oahu) without over-parameterising the models. We used Tukey's post hoc tests to run pairwise comparisons to interpret the nature of trait variation for significant effects. We used the

‘GLIMMIX’ function in SAS (SAS Institute, Cary, NC) with a binomial logit link to analyse the proportion of toads to emerge from the shelters during risk-taking trials. Animal Identity was included as a random factor in all models. We also included Sex, Mass, Arena and Trial Number as covariates in all models, as these factors might contribute to variation in behavioural traits.

We calculated adjusted repeatability (including all of the above mentioned independent variables) of behavioural traits from Trial Block One to Two using the rptR package in R (R core team 2017; rptR methods after Finger et al., 2016; Nakagawa and Schielzeth, 2010; Stoffel et al., 2017). For comparison, we calculated an intercept-only model for each behavioural trait which included Animal Identity as a random factor and excluded all other explanatory factors (adapted from Nakagawa and Schielzeth, 2010; Stoffel et al., 2017). To assess consistency of behaviours for toads in general we also calculated a global repeatability score for each behavioural trait using an intercept-only model with all sample populations grouped together. All data were checked for normality and homoscedasticity and log-transformed to meet these assumptions as required.

3. Results

3.1. Does rearing environment (common-garden versus wild) affect behaviour across repeated trials?

3.1.1. Exploration trials

3.1.1.1. Time spent moving. The amount of time that toads spent moving during exploration trials was influenced by a significant interaction effect between Rearing Environment and Trial Block (Table 1). Common-garden raised toads of both Queensland and Western Australian ancestry decreased the time that they spent moving from Trial Block One to Two (Fig. 1a). In contrast, wild-caught toads from both Queensland and Western Australia increased the time that they spent moving from Trial Block One to Two (Fig. 1b).

3.1.1.2. Rate of movement. The rate of toad movement during trials exhibited no significant interaction effect between Rearing Environment and Trial Block (Table 1). However, Rearing Environment and Trial Block had separate effects on the rate of toad movement. Overall, movement rate was higher during Trial Block One than Two (Fig. 1b).

3.1.2. Risk-taking trials

3.1.2.1. Emergence. The proportion of toads that emerged during risk-taking trials was not significantly affected by any main effects in our

model nor by the interaction between Rearing Environment and Trial Block (Table 1; Fig. 1c).

3.1.2.2. Latency to emerge. The latency of toads to emerge from the shelter during risk-taking trials was affected by a significant interaction between Rearing Environment and Trial Block (Table 1). Common-garden raised toads of Western Australian ancestry had similar latencies to emerge across Trial Blocks (Fig. 1d), whereas common-garden raised toads of Queensland ancestry reduced their latencies to emerge from Trial Block One to Two (Fig. 1d). The emergence latencies of wild-caught Australian toads did not significantly differ between Trial Blocks (Fig. 1d). Sex also had a significant effect on latency to emerge, with males quicker to emerge from shelters than were female conspecifics across both Trial Blocks (Table 1).

3.2. Does invasion history affect behaviour across repeated trials among wild-caught toads?

3.2.1. Exploration trials

3.2.1.1. Time spent moving. The amount of time that toads spent moving during exploration trials was affected by a significant interaction between Population Origin and Trial Block (Table 2). Toads from the Australian long-colonised (Queensland) population increased the time that they spent moving from Trial Block One to Two (Table 2; Fig. 2a), whereas toads from Hawai’ian populations did not significantly change their behaviour across Trial Blocks (Table 2; Fig. 2a).

3.2.1.2. Rate of movement. Movement rate was affected by a significant interaction between Population Origin and Trial Block (Table 2). Toads from long-colonised Queensland populations decreased their movement rate across Trial Blocks, whereas toads from long-colonised Hawai’ian and Australian invasion front populations did not significantly change their movement rate across Trial Blocks (Fig. 2b).

3.2.2. Risk-taking trials

3.2.2.1. Emergence. There was a significant interaction effect between Population Origin and Trial Block (Table 2). The proportion of toads from Oahu and the Big Island that emerged from the shelter during risk-taking trials was similar in Trial Block One, but diverged in Two as the proportion of Oahu toads to emerge decreased (Fig. 2c). Among Australian invasion-front toads, the proportion that emerged decreased from Trial Block One to Two, but among Australian toads from the long-colonised populations there was no change between Trial Block One and Two (Fig. 2c).

Table 1

Results from a mixed effects model testing the behavioural response of cane toads (*Rhinella marina*) from different rearing environments (captive-raised common garden [CG QLD, CG WA] and wild-caught Australian populations [Wild QLD, Wild WA]) to repeated behavioural trials testing exploration (time spent moving, movement rate) and risk-taking (proportion to emerge and emergence latency) over two Trial Blocks (separated by 12 days). Animal identity was included as a random factor. Significant effects are highlighted in bold text.

	Time spent moving (s)	Movement rate	Proportion to Emerge	Emergence latency (s)
Trial Block	$F_{1,123} = 3.76$ $P = 0.05$	$F_{1,120} = \mathbf{10.22}$ $P = \mathbf{0.002}$	$F_{1,122} = 0.03$ $P = 0.86$	$F_{1,126} = 0.04$ $P = 0.85$
Rearing Environment	$F_{3,123} = \mathbf{6.07}$ $P = \mathbf{0.0007}$	$F_{3,121} = \mathbf{4.08}$ $P = \mathbf{0.008}$	$F_{3,122} = 2.25$ $P = 0.09$	$F_{3,128} = \mathbf{5.13}$ $P = \mathbf{0.002}$
Rearing Environment*Trial Block	$F_{3,123} = \mathbf{4.78}$ $P = \mathbf{0.004}$	$F_{3,120} = 0.77$ $P = 0.51$	$F_{3,122} = 0.72$ $P = 0.54$	$F_{3,127} = \mathbf{2.77}$ $P = \mathbf{0.04}$
Sex	$F_{1,130} = 0.99$ $P = 0.32$	$F_{1,127} = 0.21$ $P = 0.65$	$F_{1,122} = 0.00$ $P = 1.00$	$F_{1,136} = \mathbf{4.52}$ $P = \mathbf{0.04}$
Mass	$F_{1,124} = 0.51$ $P = 0.48$	$F_{1,121} = 0.50$ $P = 0.48$	$F_{116,122} = 0.05$ $P = 1.00$	$F_{1,131} = 0.06$ $P = 0.81$
Trial Number	$F_{5,243} = 0.71$ $P = 0.62$	$F_{5,244} = 0.35$ $P = 0.88$	$F_{5,122} = 0.34$ $P = 0.89$	$F_{5,226} = 0.36$ $P = 0.88$
Arena Number	$F_{4,237} = 1.73$ $P = 0.14$	$F_{4,238} = 0.48$ $P = 0.75$	$F_{3,122} = 1.40$ $P = 0.24$	$F_{3,225} = 0.35$ $P = 0.79$

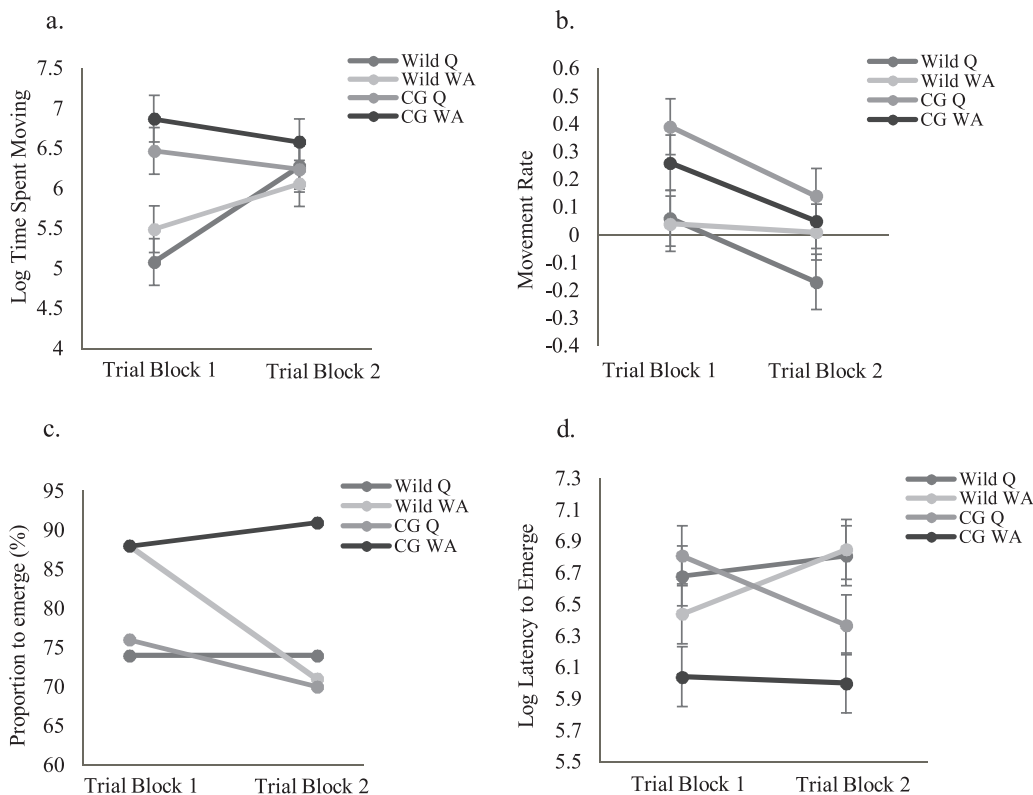


Fig. 1. In standardised behavioural trials repeated across two time periods, cane toads (*Rhinella marina*) from different Rearing Environments diverged in behavioural responses to trials designed to quantify exploration (a. Log Time Spent Moving (there was convergence in this trait in Trial Block 2); b. Movement Rate), and risk-taking (c. Proportion to Emerge and d. Latency to Emerge from shelters) behaviour. Graphs show mean values and associated standard errors (where relevant).

3.2.2.2. Latency to emerge. Emergence latency was affected by a significant interaction between population origin and Trial Block (Table 2). Toads from Oahu were slower to emerge in Trial Block Two than One, whereas all other populations did not change significantly across Trial Blocks (Fig. 2d). Sex also affected latency to emerge, with males emerging from the shelter more quickly than females (Table 2).

3.3. Repeatability of behavioural traits

3.3.1. General repeatability of cane toad behaviour (all populations combined)

The times that toads spent moving and their rates of movement were highly repeatable across Exploration Trial Blocks One and Two (Table 3). The emergence latencies of toads were moderately, but significantly, repeatable across Risk-taking Trial Blocks One and Two

(Table 3).

3.3.1.1. Rearing environment. The time that common-garden raised toads of Queensland ancestry spent moving was significantly repeatable across Trial Blocks (Table 3). Time spent moving was not significantly repeatable for common-garden toads of Western Australian ancestry, nor wild-caught toads from Queensland or Western Australia (Table 3). Similarly, the rate of toad movement was not significantly repeatable across Trial Blocks for any populations (Table 3). The emergence latencies of both wild-caught Queensland and wild-caught Western Australian populations were moderately and significantly repeatable across Trial Blocks (Table 3), whereas latency to emerge was not significantly repeatable across Trial Blocks for common-garden-raised toads of either Queensland or Western Australian ancestry (Table 3).

Table 2

Results from a mixed effects model testing the behavioural response of cane toads (*Rhinella marina*) from different populations of origin (wild-caught toads from long-colonised Queensland, Australia and Hawai'i [Oahu, Big Island], and invasion-front populations from Western Australia) to repeated behavioural trials testing exploration (time spent moving, movement rate) and risk-taking (proportion to emerge and emergence latency) over two Trial Blocks (separated by 12 days). Animal Identity was included as a random factor. Significant effects are highlighted in bold text.

	Time spent moving (s)	Movement rate	Proportion to Emerge	Emergence latency (s)
Trial Block	$F_{1,170} = 7.18$ $P = 0.008$	$F_{1,153} = 0.96$ $P = 0.33$	$F_{1,172} = 0.28$ $P = 0.60$	$F_{1,179} = 7.52$ $P = 0.007$
Population Origin	$F_{3,169} = 282.4$ $P < 0.0001$	$F_{3,152} = 18.3$ $P < 0.0001$	$F_{2,172} = 0.86$ $P = 0.43$	$F_{3,179} = 7.00$ $P = 0.0002$
Population Origin* Trial Block	$F_{3,168} = 8.03$ $P < 0.0001$	$F_{3,151} = 3.28$ $P = 0.02$	$F_{3,172} = 4.85$ $P = 0.003$	$F_{3,181} = 5.10$ $P = 0.002$
Sex	$F_{1,171} = 3.90$ $P = 0.05$	$F_{1,154} = 0.010$ $P = 0.92$	$F_{1,172} = 0.16$ $P = 0.69$	$F_{1,183} = 3.01$ $P = 0.08$
Mass	$F_{1,172} = 0.24$ $P = 0.63$	$F_{1,154} = 0.79$ $P = 0.37$	$F_{155,172} = 0.11$ $P = 1.00$	$F_{1,189} = 0.34$ $P = 0.22$
Trial Number	$F_{5,351} = 0.59$ $P = 0.71$	$F_{5,351} = 1.31$ $P = 0.26$	$F_{5,172} = 0.63$ $P = 0.68$	$F_{5,298} = 0.34$ $P = 0.89$
Arena Number	$F_{4,350} = 0.94$ $P = 0.44$	$F_{4,351} = 0.61$ $P = 0.65$	$F_{3,182} = 4.23$ $P = 0.006$	$F_{3,287} = 1.61$ $P = 0.19$

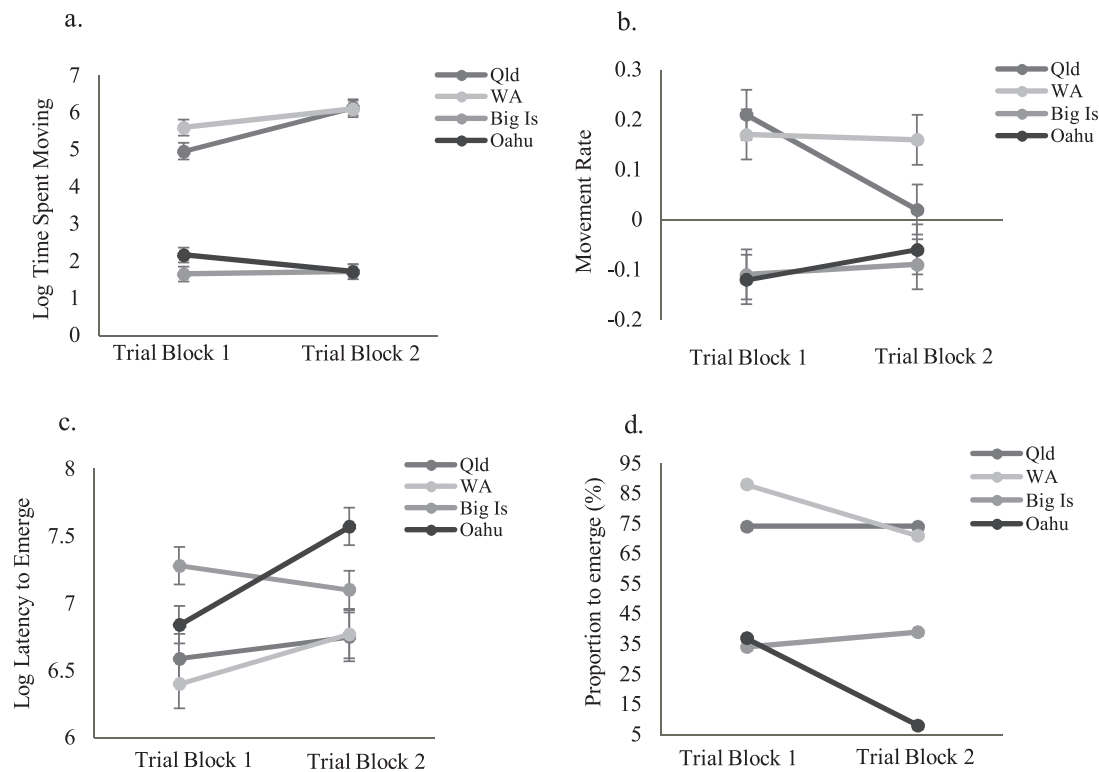


Fig. 2. In standardised behavioural trials repeated across two time periods, cane toads (*Rhynella marina*) from different Populations of Origin diverged in behavioural responses to trials designed to quantify exploration (a. Log Time Spent Moving; b. Movement Rate), and risk-taking (c. Proportion to Emerge and d. Latency to Emerge from shelters) behaviour. Graphs show mean values and associated standard errors (where relevant).

3.3.1.2. Population origin. The time that toads spent moving and their rate of movement were not significantly repeatable across Trial Blocks for any populations (Table 3). The latency to emerge was moderately repeatable across Trial Blocks for Australian toads, but not for Hawai'ian toads (Table 3).

4. Discussion

The behavioural responses of cane toads to identical trials across two time periods were complex and showed few consistent patterns based on rearing environment or population of origin. The most consistent pattern involved the impact of rearing environment on time spent moving. Toads that had been raised in captivity exhibited high movement rates in both Trial Blocks, presumably because they were already adjusted to captivity, whereas wild-caught toads from the same populations were relatively immobile in Trial Block One but more active in Trial Block Two (as they became more familiar with captivity; see Fig. 1a). Within the subgroups of toads, an individual's behaviour was not significantly repeatable across trials for most traits; hence, behavioural responses at Trial Block Two could not be predicted by behavioural responses at Trial Block One. Below, we consider possible explanations for the inconsistency and unpredictability of behavioural responses in cane toads and provide a cautionary note for future behavioural studies.

First, we examined how two factors, Rearing Environment and Population of Origin, affected behavioural responses across repeated trials. However, it is likely that behavioural responses are affected by multiple factors such as developmental and life history, personality, sex, population density, and whether an individual is from a wild or urban environment (Ellenberg et al., 2009; Ensminger and Westneat, 2012; Oosten et al., 2010). For example, in house sparrows (*Passer domesticus*), males feed sooner in the presence of novel objects and human disturbance than do females (Ensminger and Westneat, 2012) and urban sparrows habituate to human disturbance sooner than do

rural sparrows (Vincze et al., 2016). In this study, we found that Hawai'ian toads that share the same invasion history and were from similar environments but on different islands differed in habituation responses, suggesting that even populations with similar experience may diverge in unpredictable ways.

Second, the lack of consistency in behavioural responses across trials suggests high levels of behavioural plasticity. Behavioural plasticity is an adaptive response to novel environments in many taxa (Wright et al., 2010) and may enable invasive species to rapidly shift their behaviour to exploit novel environments and resources (Chevin et al., 2013). Many phenotypic traits (including behaviour) exhibit high levels of plasticity in anurans (Hossie and Murray, 2012). As ectotherms, anurans are strongly influenced by their environment (Bell et al., 2009; Ducatez et al., 2016) and factors experienced during development such as population density, food availability, predation risk and temperature can affect phenotypic traits (e.g. Bell and Sih, 2007; Hagman and Shine, 2009; Indermaur et al., 2010; Van Buskirk and Arioli, 2005). We do not have data on the developmental conditions experienced by the wild-caught toads used in this study, and hence are unable to determine whether this factor may have contributed to behavioural plasticity. However, previous research has revealed that growth rates and development of larval cane toads are affected by larval rearing densities, but with differing response patterns in individuals from long-colonised versus invasion-front populations (Ducatez et al., 2016). More directly, our comparisons of wild-caught versus common-garden-raised toads from the same populations revealed strong differences in behavioural responses to repeated trials, indicating a role for phenotypic plasticity in this respect.

Finally, the complex and unpredictable responses to repeated trials found in this study serve as a cautionary tale for using single laboratory-based behavioural trials to quantify behavioural traits. Several studies have raised similar concerns and have provided potential solutions to this problem including multiple repeated trials and large sample sizes that enable the use of random regressions for reaction norm analysis (to

Table 3
Repeatability results from linear mixed-effect models examining if behavioural traits of cane toads (*Rhinella marina*) differed across Trial Blocks 1 and 2 as a function of Rearing Environment (captive-raised common garden [CG QLD, CG WA] and wild-caught Australian populations [Wild QLD, Wild WA] or Population of Origin (wild-caught toads from long-colonised Queensland [Wild QLD] and Hawai'i [HW Oahu, HW Big Island]), and invasion-front populations from Western Australia). Repeatabilities are shown for intercept only models (R_{ADJ}) and full models including the fixed factors of Rearing Environment and Population of Origin, Trial Block, and the interactions Trial Block* Rearing Environment and Trial Block* Population Origin, Sex and Mass as covariates, and Animal Identity as the random effect (R_{ADJ}) (Nakagawa and Schielzeth, 2010). Significant repeatabilities are highlighted in bold text.

	CG QLD			CG WA			Wild QLD			Wild WA			HW Oahu			HW Big Island			Populations Combined (Global R_{ADJ})		
	R	95% CI	P	R	95% CI	P	R	95% CI	P	R	95% CI	P	R	95% CI	P	R	95% CI	P	R	95% CI	P
Time spent moving	0.43	0.27	0.81	0.01	0	0.6	1	0.001	0	0.61	1	0.07	0	0.44	0.41	0.07	0	0.44	0.41	0.80	0.001
Rate of Movement	0.49	0.2	0.72	0.001	0	0.32	1	0.03	0	0.37	0.46	0.005	0	0.25	1	0.09	0	0.32	0.27	0.69	0.80
Latency to emerge	0.001	0	0.56	0.5	0	0.41	1	0.59	0.5	0.04	0	0	0	0	1	0	0	0	1	0.87	0.92
	0	0	0	1	0.1	0	0.4	0.38	0.48	0.15	0	0.007	0	0.26	1	0.05	0	0.31	0.38	0.90	0.001
	0.12	0	0.62	0.28	0.16	0	0.63	0.2	0.49	0.81	0.001	0	0	0.33	0.5	0.23	0.05	0.51	0.13	0.14	0.37
	0.15	0	0.44	0.22	0.23	0	0.51	0.11	0.5	0.8	0.001	0	0	0	1	0.21	0	0.45	0.05	0.26	0.001

allow for habituation effects on repeatability) (Beckmann and Biro, 2013; Biro and Stamps, 2015; Dingemanse and Wolf, 2010; Réale et al., 2007; van de Pol, 2012). Although large sample sizes and data-hungry analyses such as random regressions may be ideal methods for testing consistency in behavioural traits, these methods are not practicable for some study systems. In these cases, repeated trials across different time periods may confound otherwise meaningful patterns found at one time point due to habituation responses to captivity (Martin and Réale, 2008). An alternative method may be trait validation, where an individual is measured for the same behavioural type across different trial contexts (e.g., testing for correlations of 'bold' behavioural traits in risk-taking and exploration trials), thus reducing the confounding effects of habituation to captivity over time (Burns and Rodd, 2008; Carter et al., 2013).

In the current study, we found that Rearing Environment and Population Origin had unpredictable and complex effects on the repeatability of cane toad behavioural traits across Trial Blocks One and Two. Indeed, most behavioural traits were not repeatable when Rearing Environment and Population Origin were included in repeatability models. However, we found that all of the behavioural traits tested were repeatable across Trial Blocks more generally, when all sample populations were grouped together, indicating that overall, individual toads performed in a consistent manner across replicate trials of our behavioural assays. Furthermore, previous studies using samples of toads from across the Australian invasion range collected in different years (2014, 2016) have revealed that toads from invasion-front populations consistently exhibit higher levels of exploration and risk-taking than do conspecifics from long-colonised areas (Gruber et al., 2017a,c). Behavioural trials using common-garden raised toads revealed the same pattern of geographical divergence in behaviour, suggesting a genetic component to these traits (Gruber et al., 2017c). Although toad behaviour can be highly plastic, leading to unpredictable behavioural responses across time, there nonetheless appears to be some broad-scale consistency in the rank order of population-level behavioural types (i.e., bold at the range-front versus shy at the range-core) across the invasion range.

Ethics

This research was conducted under approval no. 2013/5805 from the University of Sydney Animal Care and Ethics Committee.

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Competing interests

We have no competing interests.

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